Spring Professional 2V0-72.22 — Study Checklist

Certification-focused checklist, with notes for studying while using Spring Boot 4.1.

Part 1 — Spring Core & Dependency Injection ■■■■■

■ IoC / Dependency Injection

■ ApplicationContext

■ BeanFactory

■ @Component, @Service, @Repository, @Controller

■ @Configuration and @Bean

■ Constructor, setter and field injection

■ @Autowired

■ @Qualifier

■ @Primary

■ Multiple beans of the same type

■ Dependency resolution

■ Circular dependencies

■ Bean definition vs bean instance

Part 2 — Bean Scope & Lifecycle ■■■■■

■ Singleton scope

■ Prototype scope

■ Request / session / application / WebSocket scopes

■ @Scope

■ @PostConstruct and @PreDestroy

■ InitializingBean / DisposableBean

■ Bean lifecycle

■ Singleton vs prototype

■ Prototype injected into singleton

■ ObjectProvider / Provider

Part 3 — Spring Configuration ■■■■■

■ Java configuration

■ XML configuration concepts

■ @Configuration / @Bean

■ @Import / @ComponentScan

■ Component scanning

■ Bean naming

■ @Value

■ Property sources

■ Environment

■ Profiles / @Profile

■ @PropertySource

■ application.properties / application.yml

■ External configuration

■ Property precedence

■ SpEL

Part 4 — Spring Boot ■■■■■

■ @SpringBootApplication

■ @EnableAutoConfiguration

■ @ComponentScan

■ Starters

■ Auto-configuration

■ Conditional configuration

■ @ConditionalOnMissingBean / @ConditionalOnBean / @ConditionalOnProperty

■ Embedded server

■ Boot configuration

■ Profiles

■ Actuator

■ Externalized configuration

Part 5 — AOP ■■■■

■ Aspect, join point, pointcut, advice

■ Before / after / after-returning / after-throwing / around advice

■ @Aspect / @Pointcut / @Before / @After / @Around

■ Proxy-based AOP

■ JDK dynamic proxy

■ CGLIB proxy

■ Self-invocation problem

Part 6 — Spring Data / JPA ■■■■■

■ Repository / CrudRepository / PagingAndSortingRepository / JpaRepository

■ Derived query methods

■ @Query, JPQL and native queries

■ Pagination and sorting

■ @Entity / @Id / @GeneratedValue

■ Entity relationships

■ Lazy vs eager loading

■ Entity lifecycle

■ Persistence context

■ First-level cache

■ Dirty checking

Part 7 — Transactions ■■■■■

■ @Transactional

■ Transaction boundaries

■ Propagation

■ Isolation

■ Read-only transactions

■ Rollback

■ Runtime vs checked exceptions

■ Transaction proxies

■ Self-invocation

■ REQUIRED / REQUIRES_NEW / SUPPORTS / NOT_SUPPORTED / MANDATORY / NEVER / NESTED

Part 8 — Spring MVC / REST ■■■■■

■ @Controller / @RestController

■ @RequestMapping and HTTP method mappings

■ @PathVariable / @RequestParam

■ @RequestBody / @RequestHeader / @CookieValue

■ ResponseEntity

■ HTTP status codes

■ JSON conversion

■ Message converters

■ Content negotiation

■ @Valid / @Validated

■ Validation errors

■ @ExceptionHandler / @ControllerAdvice / @RestControllerAdvice

Part 9 — Spring Security ■■■■

■ Authentication

■ Authorization

■ Principal

■ Granted authorities

■ Roles

■ Password encoding

■ PasswordEncoder / BCrypt

■ Security filter chain

■ CSRF / CORS

■ Session management

■ Basic authentication

■ Form login

Part 10 — Testing ■■■■■

■ Unit vs integration testing

■ JUnit

■ Mockito

■ @SpringBootTest

■ Application context testing

■ Mocking concepts

■ MockMvc

■ MVC slice testing

■ Repository testing

■ Test profiles

■ Test configuration

■ Test transactions

Part 11 — JDBC ■■■■

■ JdbcTemplate

■ NamedParameterJdbcTemplate

■ RowMapper

■ DataSource

■ JDBC transactions

■ Spring exception translation

■ DataAccessException

Part 12 — Spring Messaging / Events ■■■

■ Application events

■ ApplicationEvent

■ ApplicationEventPublisher

■ @EventListener

■ Synchronous event handling

■ Asynchronous event handling concepts

Part 13 — Actuator ■■■

■ /actuator/health

■ /actuator/info

■ /actuator/metrics

■ /actuator/env

■ /actuator/beans

■ /actuator/mappings

■ Endpoint exposure

■ Health indicators

■ Metrics

Part 14 — Version Differences

■ ■ Study normally: DI, scopes, lifecycle, AOP, transactions, Spring Data, JPA, MVC, REST, validation, security concepts, testing, auto-configuration

■ ■ Be aware of: javax.* vs jakarta.*, Spring Security configuration, Boot testing APIs, configuration properties, security DSL, repository interfaces

■ ■ Low priority for certification: Spring Boot 4-specific features, Spring Framework 7-specific features, Boot 4 / Framework 7 migration details

Highest-Priority Topics

■ 1. Spring Core / DI

■ 2. Bean scopes & lifecycle

■ 3. Spring Boot / auto-configuration

■ 4. Spring Data JPA

■ 5. Transactions

■ 6. Spring MVC / REST

■ 7. Testing

■ 8. Spring Security

■ 9. AOP

■ 10. Configuration / Profiles

■ 11. JDBC

■ 12. Actuator / Events